

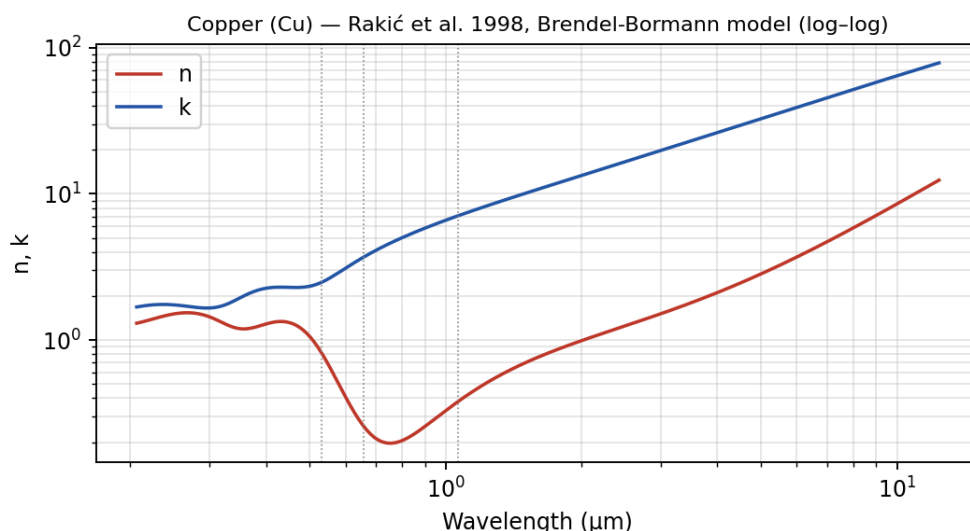
Optical constants of METALS — Copper (Cu)

Rakić et al. 1998: Brendel-Bormann model; n, k 0.2066–12.40 μm

Source: <https://refractiveindex.info/?shelf=3d&book=metals&page=copper> · Database record: database/data/main/Cu/nk/Rakić-BB.yml (public domain, CC0 1.0)

이 페이지는 "3D artists" 선반의 구리 항목으로, 실제 데이터는 MAIN 선반의 Cu → Rakić et al. 1998 (Brendel-Bormann model) 레코드와 동일함. 측정값 표가 아니라 여러 실험 데이터를 BB 모델에 피팅해 계산한 매끈한 곡선임.

Complex refractive index ($n + ik$)



[재현 그림] 데이터베이스 레코드의 표 값으로 다시 그린 n, k 곡선 (원 페이지의 대화형 그래프에 해당). 점선: 532, 658, 1064 nm.

페이지 기본 표시값 ($\lambda = 0.517 \mu\text{m}$): $n = 0.92438$, $k = 2.4023$ · $\epsilon_1 = -4.9164$, $\epsilon_2 = 4.4412$ · $\alpha = 5.8390 \times 10^5 \text{ cm}^{-1}$ · $\text{dn/d}\lambda = -7.8325 \mu\text{m}^{-1}$ · group index $n_g = 4.9738$ · 수직 입사 반사율 $R = 0.60972$ · Brewster 각 $\theta_B = 68.769^\circ$.

Comments

Fit of experimental data from several sources to Brendel-Bormann (BB) model

References

A. D. Rakić, A. B. Djurišić, J. M. Elazar, M. L. Majewski. Optical properties of metallic films for vertical-cavity optoelectronic devices. *Appl. Opt.* **37**, 5271–5283 (1998). <https://doi.org/10.1364/AO.37.005271>
Calculation script (Python): github.com/polyanskiy/refractiveindex.info-scripts — Rakić 1998 - Cu (BB model).py

주요 레이저 파장에서의 값 (계산)

λ (nm)	n	k	α (cm ⁻¹)	1/ α (nm)	R (이 데이터)	R (McPeak 2015, 비교)
532	0.8076	2.4886	5.878e+05	17.0	0.6585	0.6413
658	0.2576	3.6896	7.046e+05	14.2	0.9322	0.9745
1064	0.3786	7.0660	8.345e+05	12.0	0.9708	0.9908

원문에 없는 계산값. n, k는 아래 표에서 선형 보간, $\alpha = 4\pi k/\lambda$, $R = [(n-1)^2 + k^2] / [(n+1)^2 + k^2]$ (수직 입사, 벌크). 마지막 열은 같은 방식으로 McPeak et al. 2015 데이터에서 구한 값.

Data (tabulated n, k)

λ (μm)	n	k	λ (μm)	n	k	λ (μm)	n	k
2.0664e-01	1.3031e+00	1.6826e+00	8.2014e-01	2.1249e-01	5.1740e+00	3.2551e+00	1.6554e+00	2.1442e+01
2.1094e-01	1.3220e+00	1.7001e+00	8.3719e-01	2.2024e-01	5.3164e+00	3.3228e+00	1.6941e+00	2.1878e+01
2.1532e-01	1.3435e+00	1.7160e+00	8.5459e-01	2.2924e-01	5.4597e+00	3.3918e+00	1.7341e+00	2.2323e+01
2.1980e-01	1.3669e+00	1.7294e+00	8.7236e-01	2.3937e-01	5.6040e+00	3.4623e+00	1.7755e+00	2.2777e+01
2.2437e-01	1.3916e+00	1.7398e+00	8.9049e-01	2.5052e-01	5.7494e+00	3.5343e+00	1.8183e+00	2.3241e+01
2.2903e-01	1.4167e+00	1.7469e+00	9.0900e-01	2.6262e-01	5.8960e+00	3.6078e+00	1.8627e+00	2.3714e+01
2.3379e-01	1.4414e+00	1.7505e+00	9.2790e-01	2.7557e-01	6.0439e+00	3.6828e+00	1.9086e+00	2.4197e+01
2.3865e-01	1.4648e+00	1.7507e+00	9.4719e-01	2.8931e-01	6.1931e+00	3.7593e+00	1.9561e+00	2.4690e+01
2.4361e-01	1.4861e+00	1.7474e+00	9.6688e-01	3.0376e-01	6.3438e+00	3.8375e+00	2.0054e+00	2.5193e+01
2.4868e-01	1.5045e+00	1.7411e+00	9.8698e-01	3.1885e-01	6.4959e+00	3.9173e+00	2.0565e+00	2.5706e+01
2.5385e-01	1.5194e+00	1.7321e+00	1.0075e+00	3.3454e-01	6.6497e+00	3.9987e+00	2.1095e+00	2.6230e+01
2.5912e-01	1.5299e+00	1.7210e+00	1.0284e+00	3.5076e-01	6.8052e+00	4.0818e+00	2.1645e+00	2.6765e+01
2.6451e-01	1.5355e+00	1.7083e+00	1.0498e+00	3.6747e-01	6.9624e+00	4.1667e+00	2.2215e+00	2.7311e+01
2.7001e-01	1.5356e+00	1.6950e+00	1.0716e+00	3.8461e-01	7.1215e+00	4.2533e+00	2.2807e+00	2.7868e+01
2.7562e-01	1.5297e+00	1.6818e+00	1.0939e+00	4.0215e-01	7.2825e+00	4.3417e+00	2.3422e+00	2.8436e+01
2.8135e-01	1.5175e+00	1.6700e+00	1.1167e+00	4.2004e-01	7.4456e+00	4.4319e+00	2.4060e+00	2.9016e+01
2.8720e-01	1.4987e+00	1.6607e+00	1.1399e+00	4.3824e-01	7.6108e+00	4.5241e+00	2.4723e+00	2.9608e+01
2.9317e-01	1.4736e+00	1.6554e+00	1.1636e+00	4.5673e-01	7.7782e+00	4.6181e+00	2.5412e+00	3.0211e+01
2.9926e-01	1.4425e+00	1.6557e+00	1.1878e+00	4.7546e-01	7.9480e+00	4.7141e+00	2.6127e+00	3.0828e+01
3.0548e-01	1.4064e+00	1.6633e+00	1.2124e+00	4.9440e-01	8.1202e+00	4.8121e+00	2.6870e+00	3.1456e+01
3.1183e-01	1.3668e+00	1.6796e+00	1.2376e+00	5.1354e-01	8.2949e+00	4.9122e+00	2.7643e+00	3.2098e+01
3.1832e-01	1.3258e+00	1.7058e+00	1.2634e+00	5.3284e-01	8.4722e+00	5.0143e+00	2.8446e+00	3.2752e+01
3.2493e-01	1.2862e+00	1.7422e+00	1.2896e+00	5.5229e-01	8.6522e+00	5.1185e+00	2.9280e+00	3.3420e+01
3.3169e-01	1.2507e+00	1.7884e+00	1.3164e+00	5.7187e-01	8.8350e+00	5.2249e+00	3.0148e+00	3.4101e+01
3.3858e-01	1.2220e+00	1.8427e+00	1.3438e+00	5.9155e-01	9.0208e+00	5.3335e+00	3.1050e+00	3.4796e+01
3.4562e-01	1.2020e+00	1.9028e+00	1.3717e+00	6.1134e-01	9.2095e+00	5.4444e+00	3.1988e+00	3.5505e+01
3.5281e-01	1.1918e+00	1.9658e+00	1.4003e+00	6.3121e-01	9.4014e+00	5.5576e+00	3.2963e+00	3.6228e+01
3.6014e-01	1.1915e+00	2.0285e+00	1.4294e+00	6.5116e-01	9.5966e+00	5.6731e+00	3.3977e+00	3.6966e+01
3.6763e-01	1.2001e+00	2.0881e+00	1.4591e+00	6.7118e-01	9.7951e+00	5.7910e+00	3.5031e+00	3.7718e+01
3.7527e-01	1.2161e+00	2.1422e+00	1.4894e+00	6.9127e-01	9.9970e+00	5.9114e+00	3.6128e+00	3.8486e+01
3.8307e-01	1.2373e+00	2.1891e+00	1.5204e+00	7.1142e-01	1.0203e+01	6.0343e+00	3.7269e+00	3.9269e+01
3.9103e-01	1.2613e+00	2.2277e+00	1.5520e+00	7.3165e-01	1.0412e+01	6.1597e+00	3.8455e+00	4.0067e+01
3.9916e-01	1.2856e+00	2.2575e+00	1.5842e+00	7.5195e-01	1.0625e+01	6.2878e+00	3.9689e+00	4.0881e+01
4.0746e-01	1.3077e+00	2.2785e+00	1.6172e+00	7.7232e-01	1.0842e+01	6.4185e+00	4.0972e+00	4.1712e+01
4.1593e-01	1.3253e+00	2.2916e+00	1.6508e+00	7.9277e-01	1.1063e+01	6.5519e+00	4.2306e+00	4.2559e+01

4.2458e-01	1.3363e+00	2.2977e+00	1.6851e+00	8.1332e-01	1.1288e+01	6.6881e+00	4.3695e+00	4.3423e+01
4.3340e-01	1.3389e+00	2.2984e+00	1.7201e+00	8.3397e-01	1.1517e+01	6.8271e+00	4.5138e+00	4.4303e+01
4.4241e-01	1.3316e+00	2.2956e+00	1.7559e+00	8.5474e-01	1.1751e+01	6.9691e+00	4.6640e+00	4.5201e+01
4.5161e-01	1.3132e+00	2.2913e+00	1.7924e+00	8.7563e-01	1.1989e+01	7.1139e+00	4.8202e+00	4.6117e+01
4.6100e-01	1.2832e+00	2.2880e+00	1.8297e+00	8.9667e-01	1.2232e+01	7.2618e+00	4.9826e+00	4.7050e+01
4.7058e-01	1.2413e+00	2.2880e+00	1.8677e+00	9.1787e-01	1.2479e+01	7.4128e+00	5.1515e+00	4.8002e+01
4.8036e-01	1.1879e+00	2.2941e+00	1.9065e+00	9.3926e-01	1.2732e+01	7.5669e+00	5.3272e+00	4.8972e+01
4.9035e-01	1.1240e+00	2.3088e+00	1.9462e+00	9.6085e-01	1.2989e+01	7.7242e+00	5.5099e+00	4.9961e+01
5.0054e-01	1.0512e+00	2.3343e+00	1.9866e+00	9.8266e-01	1.3252e+01	7.8847e+00	5.6999e+00	5.0969e+01
5.1094e-01	9.7185e-01	2.3725e+00	2.0279e+00	1.0047e+00	1.3520e+01	8.0486e+00	5.8974e+00	5.1996e+01
5.2157e-01	8.8859e-01	2.4247e+00	2.0701e+00	1.0271e+00	1.3793e+01	8.2159e+00	6.1028e+00	5.3043e+01
5.3241e-01	8.0438e-01	2.4911e+00	2.1131e+00	1.0497e+00	1.4072e+01	8.3867e+00	6.3164e+00	5.4110e+01
5.4348e-01	7.2204e-01	2.5713e+00	2.1570e+00	1.0726e+00	1.4356e+01	8.5611e+00	6.5385e+00	5.5198e+01
5.5477e-01	6.4392e-01	2.6639e+00	2.2019e+00	1.0960e+00	1.4646e+01	8.7390e+00	6.7693e+00	5.6306e+01
5.6631e-01	5.7177e-01	2.7672e+00	2.2476e+00	1.1197e+00	1.4942e+01	8.9207e+00	7.0093e+00	5.7434e+01
5.7808e-01	5.0663e-01	2.8793e+00	2.2944e+00	1.1438e+00	1.5244e+01	9.1061e+00	7.2588e+00	5.8584e+01
5.9010e-01	4.4894e-01	2.9983e+00	2.3421e+00	1.1684e+00	1.5552e+01	9.2954e+00	7.5181e+00	5.9756e+01
6.0236e-01	3.9868e-01	3.1227e+00	2.3907e+00	1.1934e+00	1.5866e+01	9.4887e+00	7.7876e+00	6.0949e+01
6.1488e-01	3.5554e-01	3.2509e+00	2.4404e+00	1.2190e+00	1.6187e+01	9.6859e+00	8.0676e+00	6.2165e+01
6.2767e-01	3.1902e-01	3.3821e+00	2.4912e+00	1.2451e+00	1.6515e+01	9.8873e+00	8.3586e+00	6.3403e+01
6.4071e-01	2.8853e-01	3.5153e+00	2.5430e+00	1.2719e+00	1.6849e+01	1.0093e+01	8.6610e+00	6.4663e+01
6.5403e-01	2.6347e-01	3.6500e+00	2.5958e+00	1.2993e+00	1.7190e+01	1.0303e+01	8.9752e+00	6.5947e+01
6.6763e-01	2.4324e-01	3.7857e+00	2.6498e+00	1.3273e+00	1.7539e+01	1.0517e+01	9.3015e+00	6.7254e+01
6.8151e-01	2.2732e-01	3.9221e+00	2.7049e+00	1.3561e+00	1.7894e+01	1.0735e+01	9.6405e+00	6.8584e+01
6.9567e-01	2.1520e-01	4.0592e+00	2.7611e+00	1.3856e+00	1.8257e+01	1.0959e+01	9.9926e+00	6.9938e+01
7.1014e-01	2.0645e-01	4.1967e+00	2.8185e+00	1.4160e+00	1.8627e+01	1.1186e+01	1.0358e+01	7.1317e+01
7.2490e-01	2.0067e-01	4.3347e+00	2.8771e+00	1.4472e+00	1.9005e+01	1.1419e+01	1.0738e+01	7.2719e+01
7.3997e-01	1.9753e-01	4.4732e+00	2.9369e+00	1.4793e+00	1.9390e+01	1.1656e+01	1.1132e+01	7.4147e+01
7.5535e-01	1.9675e-01	4.6121e+00	2.9979e+00	1.5124e+00	1.9784e+01	1.1899e+01	1.1541e+01	7.5599e+01
7.7105e-01	1.9806e-01	4.7516e+00	3.0602e+00	1.5465e+00	2.0186e+01	1.2146e+01	1.1966e+01	7.7077e+01
7.8708e-01	2.0124e-01	4.8917e+00	3.1239e+00	1.5817e+00	2.0596e+01	1.2398e+01	1.2406e+01	7.8579e+01
8.0344e-01	2.0610e-01	5.0325e+00	3.1888e+00	1.6179e+00	2.1014e+01			

데이터베이스 레코드 원문 그대로 200행 (브라우저에서 계산한 해시와 대조해 전 행 일치 확인).